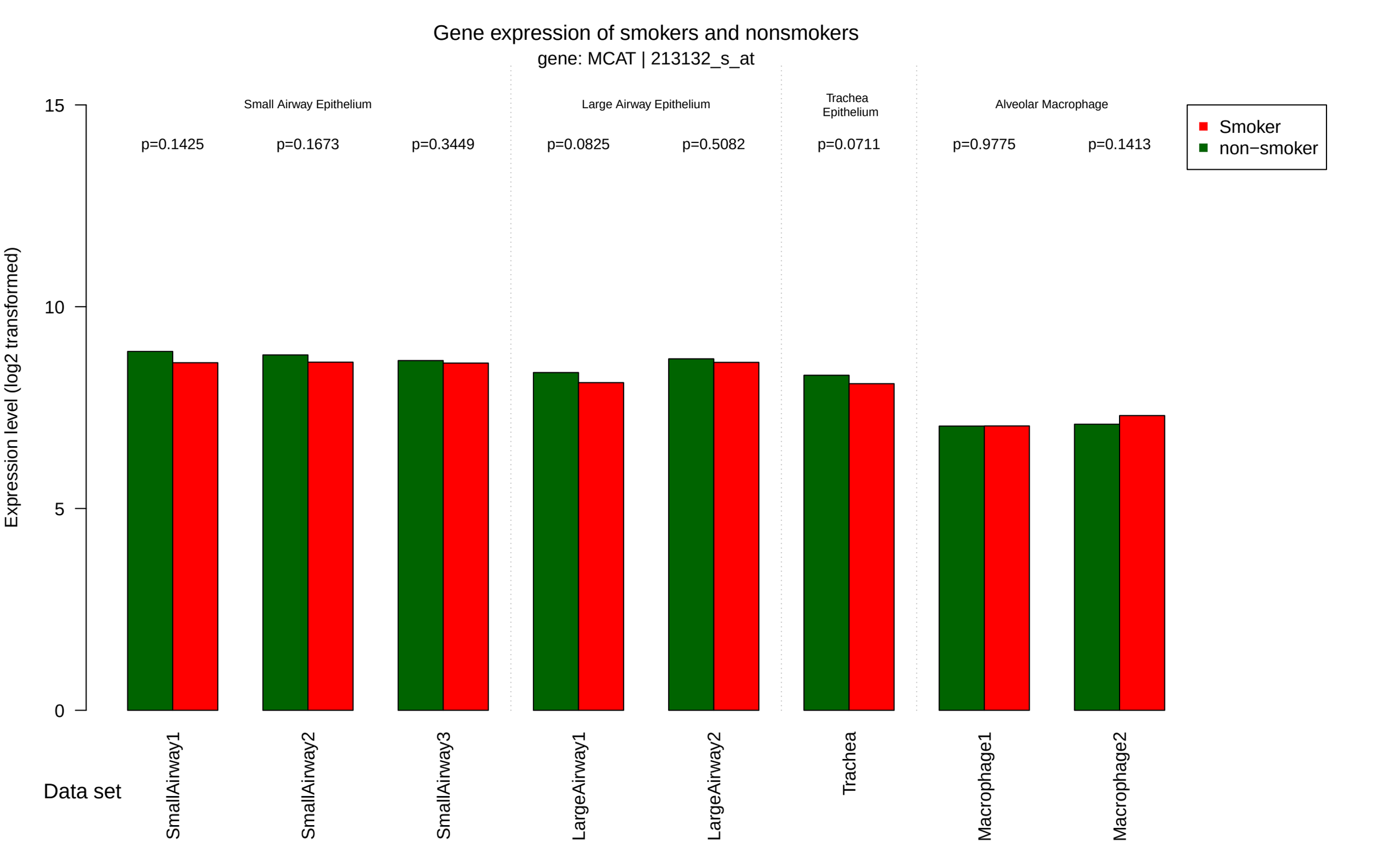
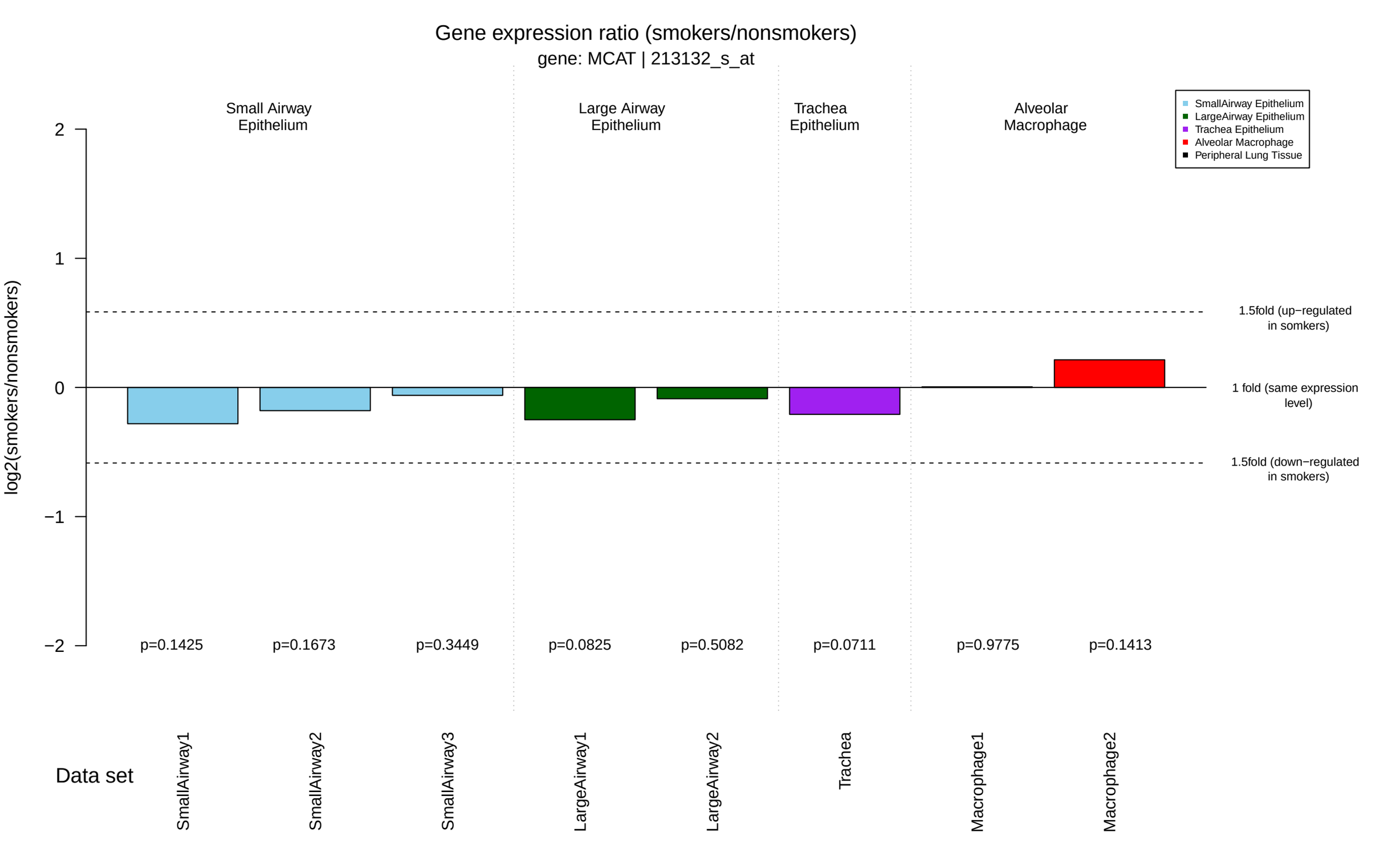
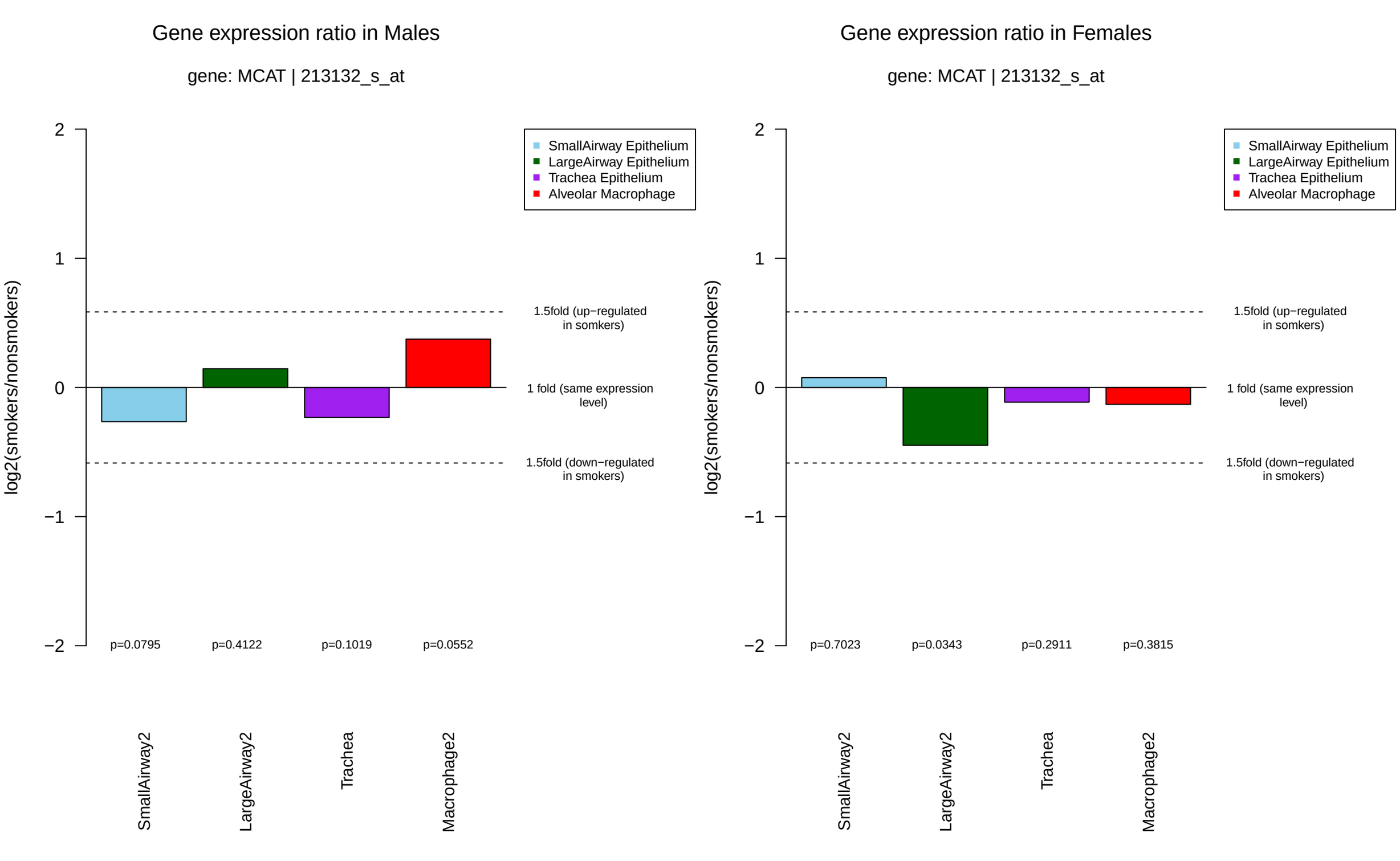
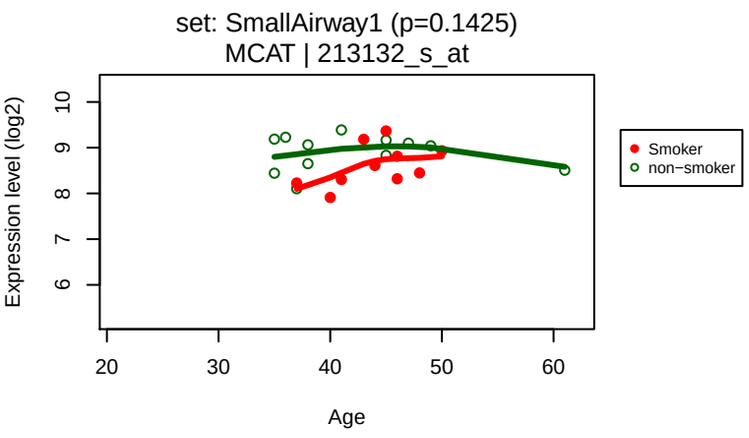
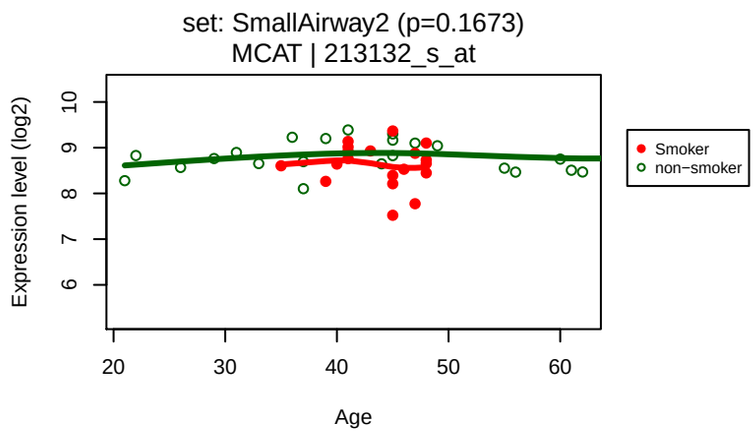
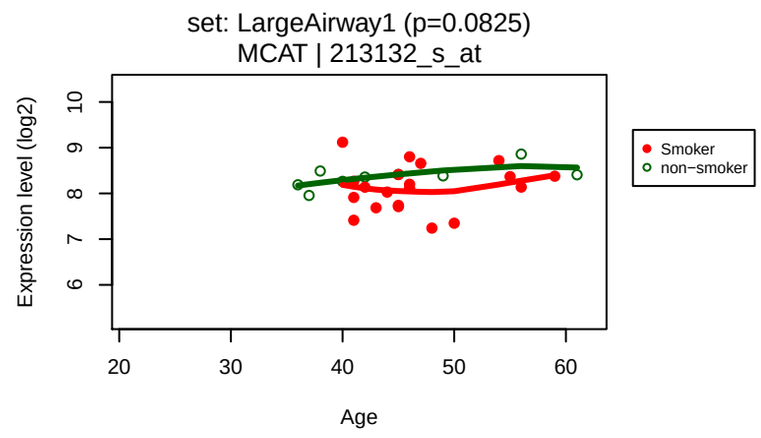

Gene expression of smokers and nonsmokers
gene: MCAT | 213132_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1425
p=0.1673
p=0.3449
p=0.0825
p=0.5082
p=0.0711
p=0.9775
p=0.1413
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

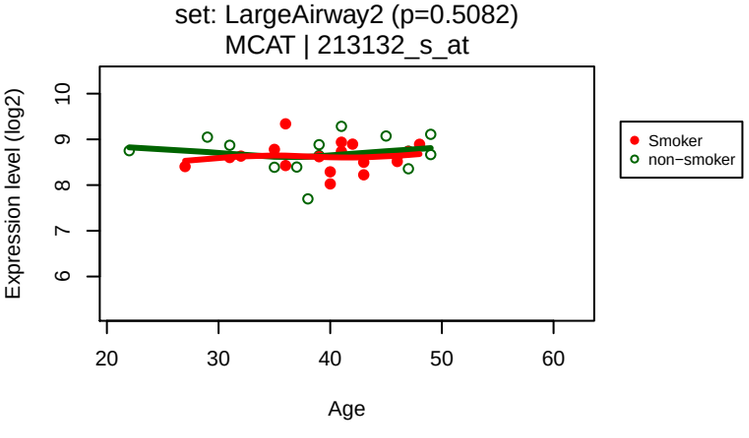

Gene expression ratio (smokers/nonsmokers)
gene: MCAT | 213132_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1425
p=0.1673
p=0.3449
p=0.0825
p=0.5082
p=0.0711
p=0.9775
p=0.1413
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

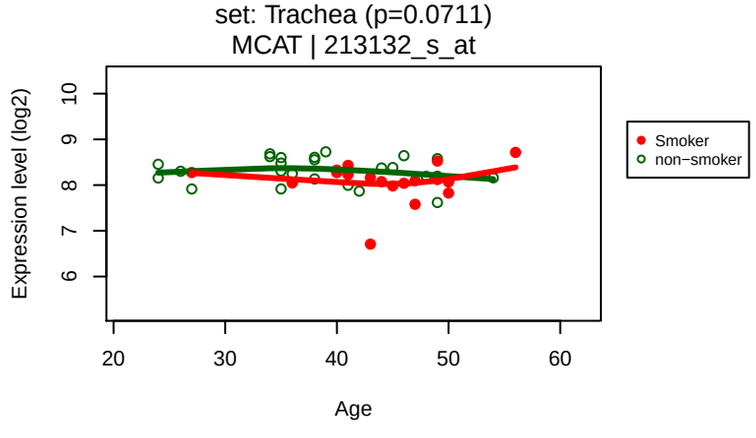

Gene expression ratio in Males
gene: MCAT | 213132_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0795
p=0.4122
p=0.1019
p=0.0552
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MCAT | 213132_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7023
p=0.0343
p=0.2911
p=0.3815
SmallAirway2
LargeAirway2
Trachea
Macrophage2

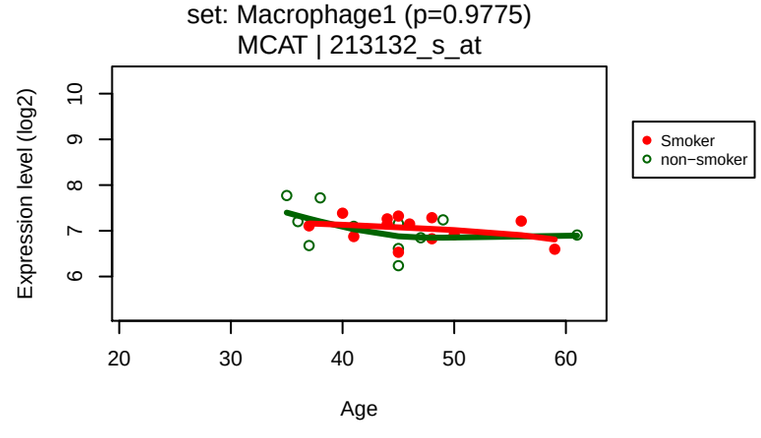

set: SmallAirway1 (p=0.1425)
MCAT | 213132_s_at
Expression level (log2)
Age
Smoker
non−smoker

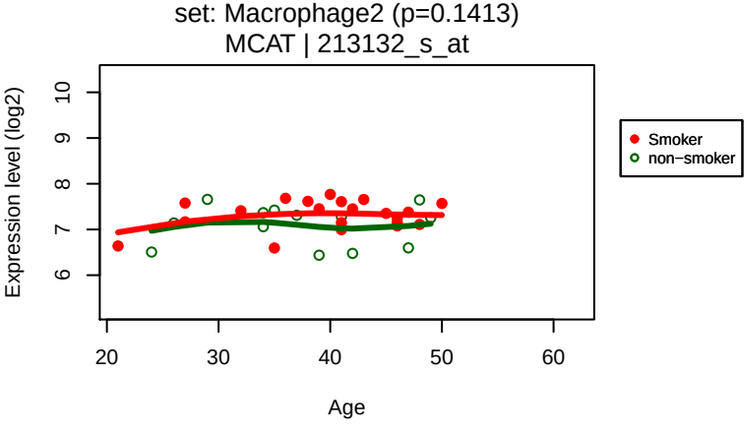

set: SmallAirway2 (p=0.1673)
MCAT | 213132_s_at
Expression level (log2)
Age
Smoker
non−smoker

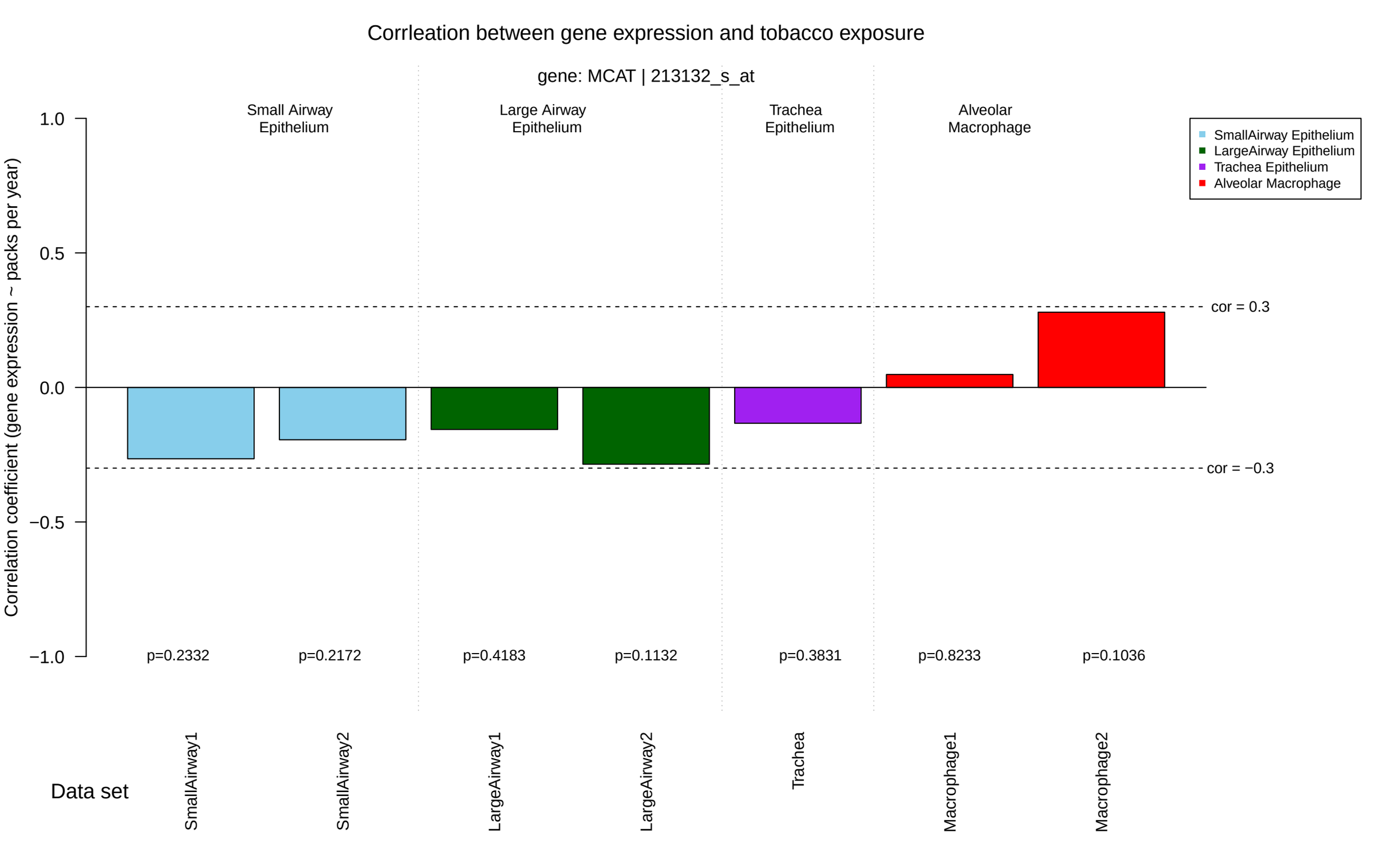

set: LargeAirway1 (p=0.0825)
MCAT | 213132_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5082)
MCAT | 213132_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0711)
MCAT | 213132_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9775)
MCAT | 213132_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1413)
MCAT | 213132_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MCAT | 213132_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2332
p=0.2172
p=0.4183
p=0.1132
p=0.3831
p=0.8233
p=0.1036
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2